

Stress-Testing Cross-Participant PAC Forecasts During 40 Hz Auditory Stimulation

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Abstract: This retrospective computational study tested whether a frontal theta-phase/40 Hz-amplitude modulation-index feature, referred to here as PAC, can be predicted across participants during 40 Hz auditory stimulation, and whether the result changes when PAC targets are computed from backward-looking EEG rather than complete event summaries. Recordings from 35 participants in OpenNeuro ds005048 were processed using seven frontal channels and split by participant into 24 training, 5 validation, and 6 held-out test participants. In an event-summary benchmark, a compact 12-feature TCN achieved held-out test R-squared of 0.558, compared with -0.025 for the 73-feature candidate set. Across five training seeds on the same fixed split, mean R-squared was 0.606 versus 0.104 for persistence. Because complete-event PAC values assigned to early windows include later samples, a stress test recomputed PAC from five-second backward-looking contexts. Unique held-out targets increased from 104 to 2,678, and adjacent target repetition fell from 96.2 percent to zero. Under this single-seed stress test, TCN R-squared decreased from 0.554 to 0.212 and did not exceed Ridge regression at 0.216. Offline controller replay increased low-PAC stimulation targeting relative to reactive thresholding, 73.8% versus 51.7%, but reduced high-PAC rest specificity, 50.7% versus 77.3%, leaving balanced alignment lower. These results treat PAC as an operational EEG feature: under the present PAC definition, target construction controlled the main conclusion more than model complexity.

Keywords: auditory stimulation, phase-amplitude coupling, EEG, temporal convolutional network, target-definition audit, retrospective replay

I. INTRODUCTION

Gamma-frequency neural activity is being studied in neurodegeneration research. In 5XFAD mice, experimentally induced 40 Hz activity reduced amyloid burden and altered microglial response [1]. Human electroencephalography (EEG) studies have also investigated non-invasive auditory gamma-frequency entrainment. In a memory-clinic cohort, Lahijanian et al. reported 40 Hz power and connectivity measures during stimulation, consistent with a stimulation-locked EEG response [2]. These findings motivate tools for characterizing neural response, but they do not establish that an adaptive auditory controller improves neural physiology or clinical outcomes.

In this paper, 40 Hz auditory stimulation is the protocol context. The modeled outcome is frontal PAC, not source-level entrainment, clinical status, or treatment response.

Many stimulation protocols use fixed timing. A fixed schedule is simple, reproducible, and easy to evaluate, but it cannot respond to within-session changes in measured brain state. Closed-loop neuromodulation research instead seeks to adjust stimulation from measured biomarkers [4]. A useful controller would need to detect or forecast low values of an operational biomarker before choosing stimulation. That objective requires a forecast that generalizes across participants, an online biomarker estimator, and a controller whose decision rule balances stimulation during low-biomarker periods against rest during high-biomarker periods. This study uses low PAC as a control target; it does not test whether low PAC marks therapeutic need or whether stimulation raises PAC.

This project separates two tasks: estimating the current biomarker and forecasting its future value. This paper isolates the temporal forecasting stage with stored PAC labels, then

stress-tests whether its conclusions hold when PAC is recomputed from backward-looking signal only. A broader prototype includes a compact raw-EEG estimator, but that estimator is not composed with the forecasting or replay validation loop reported here.

This study focuses on three questions:

1. Can a compact temporal model predict stored PAC-series entries five index steps ahead on participants excluded from training?
2. Does the result change when PAC is recomputed from backward-looking contexts instead of complete-event summaries?
3. When the forecast is used in offline replay, what sensitivity-specificity tradeoff appears relative to fixed and reactive decision rules?

The contribution is a held-out-participant PAC forecasting benchmark, a target-definition stress test, and an offline replay diagnostic showing how forecast-driven decisions trade low-PAC sensitivity against high-PAC specificity.

Feature ablation found that the compact input set worked best in the event-summary benchmark. A 12-feature representation of PAC trajectory and stimulation context generalized better than a larger 73-feature candidate representation on the fixed held-out split. The backward-looking stress test then showed a larger limitation: the TCN's advantage over a linear baseline did not persist. Controller replay showed a separate calibration tradeoff. These results narrow what must be fixed before a live system is plausible.

II. DATA AND METHODS

A. Dataset and Preprocessing

The analysis used the Brain Imaging Data Structure (BIDS)-formatted OpenNeuro dataset ds005048 version 1.0.1 [3]. The local release contains auditory-stimulation EEG recordings from 35 participants recruited from a memory clinic. Participant labels include normal cognition, mild cognitive impairment, and Alzheimer's disease classifications; two participants have missing group labels. The source study recorded 19 monopolar EEG channels at 250 Hz during 40 Hz auditory stimulation trials interleaved with rest periods [2].

Seven frontal channels were selected: Fp1, Fp2, F7, F3, Fz, F4, and F8. The source study reports Makoto-pipeline preprocessing, including a 1 Hz high-pass filter, 50 Hz line-noise removal, artifact subspace reconstruction, average rereferencing, and ICA-based component rejection [2]. Released sidecar files independently report the 1 Hz high-pass filter, 50 Hz notch filter, and average reference. The local processing stage then applied an additional 0.5-80 Hz zero-phase band-pass filter, a 50 Hz zero-phase notch filter, absolute-amplitude masking above 100 microvolts, and common-average rereferencing. Masked samples were set to zero before rereferencing. Because frontal/frontopolar gamma-band EEG is vulnerable to ocular and facial-muscle activity [6], this deterministic conditioning should not be read as a validated artifact-removal pipeline for physiological PAC. Each event period was segmented into two-second windows with a one-second hop.

B. PAC Label Construction

Theta-gamma PAC was measured using the modulation index introduced by Tort et al. [5]. For each channel, theta phase was extracted from 4-8 Hz, gamma amplitude from 38-42 Hz, and the gamma amplitude distribution was summarized over 18 theta-phase bins. The normalized divergence from a uniform distribution produced a dimensionless modulation-index value. Values were averaged across the seven selected channels to create an engineering scalar target, not a spatial connectivity analysis.

The modulation-index values are treated as operational EEG features. No surrogate correction or phase-clustering debiasing was applied, so the values should not be read as definitive evidence of physiological theta-gamma coupling, endogenous gamma activity, source-level entrainment, or therapeutic response [7].

PAC was computed over complete event periods rather than individual two-second windows because short windows provide too few theta cycles for stable phase binning. Most event periods lasted 20 or 40 seconds, with brief final rest periods also present. Each two-second window inherited the PAC value of its containing period. This stabilizes the label but has three consequences: labels repeat within periods, adjacent samples are not independent, and a PAC value assigned to an early window summarizes signal samples acquired later in that period. The present analysis is therefore retrospective; an online estimator must be evaluated before live use.

C. Temporally Ordered Forecasting Dataset

Each temporal sample used a 20-step lookahead and a target five steps into the future. With a one-second hop, this corresponds to 20 stored-series seconds of history and a target five seconds after the sequence endpoint. Sequences were constructed separately within each participant so that no sequence crossed a participant boundary. Target indexing uses no later PAC-series entry as an input, but it does not resolve the online-availability limitation of the event-period PAC summaries.

The initial representation contained 73 candidate features per time step:

- 61 spectral features
- 7 PAC-history features: current PAC, trailing means over 2, 4, 8, and 16 steps, and differences over 1 and 4 steps
- 5 stimulation-context features: stimulation state, normalized time since switch, recent stimulation fraction, and sine/cosine protocol-phase terms.

An exploratory ablation identified the 12 PAC-history and stimulation-context features as the best-performing held-out test representation in this fixed split. Because representation choice was informed by this benchmark, the result should be treated as exploratory until repeated participant-level splits or nested model selection are run. The data were divided by participant into 24 training, 5 validation, and 6 test participants. The resulting temporal dataset contained 11,160 training, 2,605 validation, and 2,678 test sequences. Normalization statistics were fit on the training split only and then applied to all splits.

D. Backward-Looking PAC Stress Test

To test how strongly the event-summary target definition influenced the result, an additional benchmark reconstructed contiguous event segments from the overlapping windows and recomputed PAC at each timestamp from a five-second backward-looking context ending at that timestamp. The benchmark used the same 20-step lookahead, five-step horizon, 12 features, participant splits, and train-only normalization. For contexts shorter than three seconds at segment boundaries, the PAC computation used 9 rather than 18 phase bins.

In this stress test, each input PAC value was computed without EEG samples after its own timestamp. However, the benchmark still used zero-phase filtering within each bounded context, and five-second PAC contexts provide limited theta-cycle support. It is therefore an information-boundary stress test that is closer to online availability than complete-event back-assignment, but it is not a validated streaming estimator.

E. Temporal Model

The selected forecaster is a lightweight temporal convolutional network (TCN), which applies temporal convolutions to ordered samples [8]. An input projection is followed by four residual depthwise-separable temporal blocks with kernel size 3 and dilations [1, 2, 4, 8]. Within the stored input sequence, left-only padding prevents later entries from entering an earlier temporal representation. Attention pooling summarizes the 20-step sequence before regression heads estimate future PAC and PAC change.

The primary forecasting study evaluated the selected 12-feature representation across five random seeds. For controller

integration, a dual-head 12-feature checkpoint was trained separately with the same hidden-64 architecture. That architecture contains 27,139 trainable parameters. Forecasting-study, stress-test, and controller-replay results are reported separately because they come from distinct retraining or replay runs.

F. Retrospective Controller Replay

The integrated checkpoint was evaluated in offline replay on all 35 recorded PAC trajectories, including the training, validation, and test splits. This replay probes controller behavior; it is not an additional held-out forecasting test. The replay compared four strategies:

1. **Fixed schedule:** replay the nominal recorded stimulation schedule.
2. **Reactive threshold:** after a 10-sample warm-up, stimulate when current PAC is more than 0.5 standard deviations below its 30-sample rolling baseline; otherwise rest.
3. **TCN predictive:** stimulate or rest when normalized forecast PAC change crosses -0.3 or $+0.3$; otherwise use a current-PAC threshold fallback with a three-step minimum hold before switching state.
4. **Alignment oracle:** stimulate below each participant's full-trajectory median PAC and rest otherwise.

The oracle uses the entire recorded trajectory and is not a causal or deployable strategy. For each participant, low PAC was defined relative to that participant's full-trajectory median. The low-PAC stimulation rate measured the fraction of low-PAC points assigned stimulation. The high-PAC rest rate measured the fraction of high-PAC points assigned rest. Balanced alignment was their mean:

$$\text{alignment} = (\text{low-PAC stimulation rate} + \text{high-PAC rest rate}) / 2.$$

The replay evaluates decision alignment against recorded PAC. It does not estimate how PAC would physiologically change if counterfactual stimulation decisions were delivered. Because the stimulation-context features are computed from the nominal recorded protocol rather than the controller's decisions, replay does not model a closed feedback loop. It is an offline controller diagnostic, not a prospective experiment.

III. RESULTS

A. Feature Selection Improved Held-Out Forecasting

Table I shows a single-seed TCN ablation on the same participant split. Adding the spectral feature set reduced held-out-participant generalization, while PAC history and stimulation context produced the strongest result.

TABLE I. ORIGINAL EXPLORATORY SINGLE-SEED FEATURE ABLATION AT A FIVE-SECOND HORIZON

Input representation	Features	Test R2
Spectral only	61	-0.420
All candidate features	73	-0.025
PAC trajectory only	7	0.344
PAC trajectory + stimulation context	12	0.558

The ablation does not prove a biological mechanism for the spectral features' failure. It shows that the larger spectral representation introduces variation that does not transfer effectively to the held-out participants in this dataset.

Across five seeds, the selected 12-feature TCN achieved mean test R-squared = 0.606, with a range from 0.558 to 0.647. At the same five-second horizon, a last-value persistence baseline achieved R-squared = 0.104.

B. Backward-Looking PAC Changed the Model-Selection Result

Figure 1 compares the event-summary benchmark with the backward-looking PAC stress test using the same TCN architecture at one random seed and one fixed split. Unique held-out targets increased from 104 to 2,678, and adjacent target repetition decreased from 96.2% to 0.0%. Persistence R-squared decreased from 0.104 to -0.897. The backward-looking target retained modest predictable structure, but the TCN did not outperform Ridge regression (0.212 versus 0.216). The event-summary result is a diagnostic of stored-series structure, not evidence that a nonlinear TCN is necessary for an online-compatible forecaster.

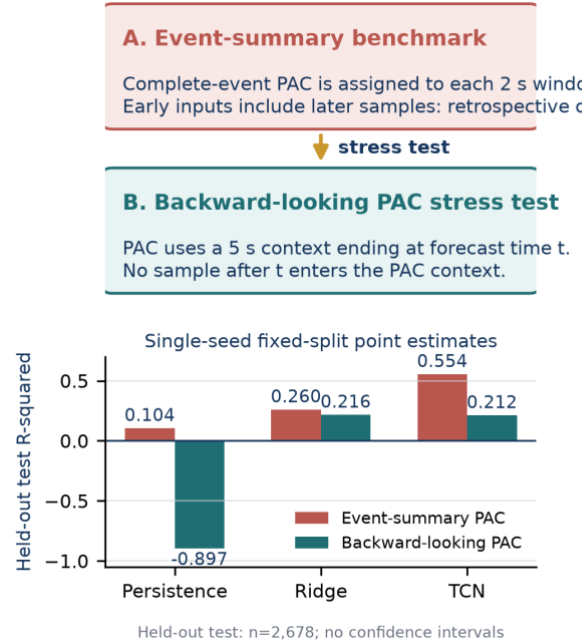


Fig. 1. *Target-definition stress test. Complete-event PAC back-assignment includes later samples in early-window inputs; backward-looking PAC restricts each context to samples available by its timestamp. Bars show single-seed fixed-split R-squared on six held-out participants (2,678 sequences); no confidence intervals are shown.*

C. Predictive Replay Increased Low-PAC Targeting but Reduced Specificity

Table II reports offline replay of the integrated 12-feature checkpoint across all 35 recorded trajectories. The predictive controller improved low-PAC stimulation targeting by 22.1 percentage points relative to reactive thresholding, assigning stimulation to more below-median PAC samples. However, high-PAC rest specificity decreased by 26.6 points, assigning

rest to fewer above-median PAC samples. This is a sensitivity-specificity tradeoff rather than a controller improvement.

TABLE II. RETROSPECTIVE CONTROLLER REPLAY ON 35 RECORDED TRAJECTORIES

Strategy	Bal. align.	Low stim.	High rest	MI gap $\times 10^{-6}$
Fixed schedule	45.0%	61.4%	28.6%	-6.55
Reactive threshold	64.5%	51.7%	77.3%	21.09
TCN predictive (12-feature)	62.2%	73.8%	50.7%	21.02
Alignment oracle	100%	100%	100%	33.36

The PAC gap is mean PAC during rest minus mean PAC during stimulation, reported in $\times 10^{-6}$ modulation-index units. A positive value indicates that stimulation decisions are concentrated more strongly in lower-PAC periods. The predictive and reactive strategies produced nearly identical PAC gaps despite different action profiles.

D. Leakage Checks and Label-Resolution Caveat

The stored temporal dataset was audited for subject overlap, future-index ordering, and normalization contamination. No participant appeared in more than one split. Every target index was exactly five steps after the sequence endpoint. Recomputed training statistics matched the stored normalizers. A reduced-model permutation test produced approximately zero validation R-squared across five independent label shuffles.

The main caveat is how PAC labels were computed and assigned. Because PAC is computed over complete event periods, 82.2% of five-step sample pairs have identical current and target PAC values. Persistence still achieves only R-squared = 0.104 on the held-out test split and R-squared = -0.272 on cross-event transitions. However, event-period back-assignment means that the stored current-PAC input is not yet a validated online feature. The forecasting results evaluate retrospective temporal structure, not an end-to-end streaming predictor.

IV. DISCUSSION

The forecasting results show that stored PAC histories and stimulation context contain information that predicts PAC in held-out participants under the event-summary label definition. Feature selection mattered within that benchmark: the compact representation transferred to unseen participants more effectively than the larger candidate set. The target definition mattered more. When PAC was recomputed from backward-looking contexts, Ridge and TCN both had positive held-out R-squared, but the nonlinear advantage disappeared. This suggests that target construction should be benchmarked before adding model complexity.

The replay separates forecast quality from controller performance. The integrated checkpoint stimulates more low-PAC periods, but it also stimulates too frequently during high-PAC periods. A future controller should tune its objective for both sensitivity and specificity, evaluate thresholds on held-out participants, feed counterfactual actions back into stimulation-context features during simulation, and replace back-assigned

PAC summaries with an online estimator. A corrected-target feature ablation should also separate PAC-only, stimulation-context-only, and combined inputs to distinguish biomarker dynamics from recorded schedule structure.

The study has seven main limitations:

1. The controller evaluation is retrospective replay, not a live intervention.
2. The replay cannot estimate physiological responses to counterfactual stimulation timing.
3. Primary PAC inputs are complete-event summaries assigned back to constituent windows and are not validated as streaming-available features.
4. The backward-looking stress test uses zero-phase filtering inside bounded contexts and is not a production streaming estimator.
5. PAC estimates may be sensitive to filtering, artifact handling, and finite-sample bias.
6. The 38-42 Hz amplitude band overlaps the auditory stimulation frequency, so PAC estimates may reflect stimulation-locked steady-state response or residual artifact rather than endogenous gamma coupling.
7. The dataset is single-site and modest in size; the current results use one fixed participant split.

Because of these limitations, this study should not make clinical or therapeutic efficacy claims. The next experiments should evaluate additional backward-looking PAC context lengths, implement a streaming-compatible PAC estimator, measure repeated participant-split performance, calibrate controller objectives using held-out participants, and then test a prospective streaming implementation.

V. CONCLUSION

In an event-summary benchmark, a 12-feature TCN predicted stored PAC-series entries on held-out participants more accurately than a larger 73-feature candidate representation. Across five training seeds on the same fixed split, mean forecasting performance reached R-squared = 0.606; this estimates initialization variability, not sensitivity to which participants are held out. A backward-looking PAC stress test then removed the repeated-target shortcut: the TCN retained positive held-out R-squared but did not exceed Ridge regression. Offline replay showed stronger low-PAC targeting without improved balanced alignment. The results support compact temporal features as a retrospective benchmark, but they also show that the PAC target definition controls the conclusion. A streaming PAC estimator, artifact sensitivity analysis, repeated participant-split validation, and a separately validated controller are required before live adaptive-stimulation testing.

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